Joongheon Kim

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Educational Backgrounds

- **University of Southern California (USC) –** Viterbi School of Engineering — *Los Angeles, California, USA*
  - Ph.D. (08/2009–08/2014) in **Computer Science**, Mark and Mary Stevens School of Computing and Artificial Intelligence
    Advisor: Prof. Andreas F. Molisch (Communications, Information, Learning & Quantum (CLIQ) Group (Electrical Engineering))
  - M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
  - M.S. (05/2012) in **Electrical Engineering**
- **Korea University –** College of Informatics — *Seoul, Republic of Korea*
  - M.S. (03/2004–02/2006) in **Computer Science and Engineering**
  - B.S. (03/1999–02/2004) in **Computer Science and Engineering**

Professional Affiliations

Full-Time

- **Korea University**, *Seoul, Republic of Korea*
  Professor (09/2019–Present), Department of Electrical and Computer Engineering
  Adjunct Professor (03/2023–Present), Department of Communications Engineering (Samsung Electronics)
  Adjunct Professor (03/2021–Present), Department of Semiconductor Engineering (SK Hynix)
  Adjunct Professor (11/2022–Present), Department of Future Science and Technology Business (Graduate School)
  Affiliated Professor (03/2025–Present), Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, *Seoul, Republic of Korea*
  Assistant Professor (03/2016–08/2019), School of Computer Science and Engineering
- **Intel Corporation**, *Santa Clara (Silicon Valley), California, USA*
  Systems Engineer (09/2013–02/2016), Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, *Los Angeles, California, USA*
  Ph.D. Student in Computer Science (08/2009–08/2014), Thomas Lord Department of Computer Science
  Annenberg Graduate Fellow (2009), 4-Year Full Scholarship for the Ph.D. Program in Computer Science
  Research Assistant (01/2011–08/2014), Communications, Information, Learning & Quantum (CLIQ) Group
  Teaching Assistant (01/2012–05/2013), Computer Science and Electrical Engineering Major Courses
- **InterDigital**, *San Diego, California, USA*
  Intern (05/2012–08/2012), Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, *Seoul, Republic of Korea*
  Research Engineer (01/2006–08/2009), Multimedia Research Lab, Seocho R&D Campus

Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, *Seoul, Republic of Korea*
  Visiting Professor (10/2025–Present), Healthcare AI Research Institute

Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, **Advisory Committee (12/2025–Present)**
- **National Research Foundation of Korea**, **Review Board (11/2024–10/2026)**

University R&D/Administrative Positions

R&D Leadership Positions

- **Korea University**, *Seoul, Republic of Korea*
  Director (07/2024–12/2031), Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
  Principal Investigator, Software Star-Lab (07/2024–12/2031), Quantum AI Empowered Second-Life Platform Technology

Administrative Positions

- **Korea University**, *Seoul, Republic of Korea*
  Associate Dean (07/2026–Present), University College
  Vice Department Chair (01/2025–08/2025), Academic Affairs, School of Electrical Engineering
  Deputy Vice President (02/2022–08/2024), Office of Academic Affairs
  Dean (06/2021–08/2023), Center for Teaching and Learning (CTL)

Committee Members

- **Korea University**, *Seoul, Republic of Korea*
  Member (12/2023–Present), Council of the Interdisciplinary Program in Quantum Information Science
  Member (03/2021–02/2027), Korea University 120th Anniversary Planning Committee
  Member (06/2021–03/2022, 09/2022–08/2023), Academic Affair Planning Committee
  Member (03/2021–08/2023), Distance Learning Management Committee
  Steering Committee Member (07/2020–06/2022), Institute of Data Science (IDS)

Awards and Honors

Research and Academic Excellence (International)

- **Top Cited Article in ETRI Journal (Wiley), among work published in 2024** 2025
- **Finalist, AAAI Student Abstract and Poster Session – Oral Presentation** 2026 (Top 35), 2023 (Top 25)
- **IEEE ICTC Best Paper Award** – *IEEE Communications Society* 2022
- **Spotlight, Oral Presentation** – *ICML Workshop on Dynamic Neural Networks (2022)* 2022
- **IEEE MMTC Best Journal Paper Award** – *IEEE Communications Society* 2021
- **IEEE ICOIN Best Paper Award** – *IEEE Computer Society* 2021
- **IEEE MMTC Outstanding Young Researcher Award** – *IEEE Communications Society* 2020
- **IEEE Systems Journal Best Paper Award** – *IEEE Systems Council* 2020
- **Next Generation and Standards (NGS) Division Recognition Award** – *Intel Corporation* Q1/2015
- **Annenberg Graduate Fellowship Award** – *University of Southern California* 2009–2013
  Awarded with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and $120,000 Stipend ($30,000/year for 4 years))
- **IEEE Seoul Section Student Paper Contest** Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- **IEEE VTS Seoul Chapter Award** – *IEEE Vehicular Technology Society* 2023, 2022, 2022, 2021, 2021, 2019

Research and Academic Excellence (Korea Regional, *since 2016*)

- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications)** – *KICS* 2025
- **Best Paper Award, The Journal of KICS** – *KICS* 2024
- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications)** – *KICS* 2023
- **Haedong Young Scholar Award** – *KICS* and *Haedong Foundation* 2018
  For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

Korea University

- **Granite Tower Best Research Award (Top 3%)**, Totally, 2 times
  – 2025 (Top 3%, top 53 professors at Korea University in 2025) 05/2026
  – 2024 (Top 3%, top 54 professors at Korea University in 2024) 05/2025
- **Best Research Achievement Award** –*Korea University, School of Electrical Engineering* 2024
- **Insung Research Grant Award** – *Korea University* 01/2023
  For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- **Granite Tower Best Teaching Award (Top 5%)**, Totally, 7 times
  – Probability and Random Process KECE209-02 (2025)
  – Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
  – Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
  – Introduction to Computers SEMI103 (2021)
  – Future Mobility Technology GEQR075 (2022)
- **Best Teaching Award (Top 20%)**, Totally, 7 times
  – Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
  – Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
  – Computer Language and Laboratory EGRN151-09 (2020)
  – Object-Oriented Programming SEMI104 (2021)

Academic and University Services (International)

- **Certificate of Appreciation** – *IEEE Future Networks, AI/ML Working Group* 08/2025
- **Certificate of Appreciation** – *IEEE/IFIP WiOpt (2024)* 10/2024
- **Best Editor Award** – *ICT Express (Elsevier)* 07/2023
- **2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter**, Awarded as a Treasure 2022
- **Appreciation Recognition** – *Daegu Gyeongbuk Institute of Science and Technology (DGIST)* 2021
- **Certificate of Appreciation** – *Department of Computer Science, University of Southern California* 2010

Academic and University Services (Korea Regional)

- **Outstanding Contribution Award**
  – The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
  – Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
  – Open Standards and ICT Association (OSIA) 2025, 2021
  – The Institute of Electronics and Information Engineers (IEIE) 2025

- **Commemorative Plaque**, *KICS Excellent Lecture* (Title: Quantum Machine Learning and Software Technologies) 2025
- **Best Research Group Award**, *KICS Research Group in Quantum Communications and Quantum Computing* (Group Leader) 2025
- **Appreciation Recognition –** *Daegu Gyeongbuk Institute of Science and Technology (DGIST)* 2021

Student Awards (International)

- **IEEE TCSC Award for Excellence in Scalable Computing** (IEEE Computer Society), Awarded to *Soohyun Park* 2025
- **Young Scientist Award** (ICSANE 2025), Awarded to *Minji Lee* 2025
- **IEEE Vehicular Technology Society Student Scholarship Award (2025)**, Awarded to *Gyu Seon Kim* 2025
- **IEEE ICDCS Best Runner-Up Poster Paper Award (2025)**, Awarded to *Seok Bin Son, Soohyun Park* 2025

R&D Projects

University-Wide/Center/Flagship Projects

- **(Center Director, PI) Net-Zero CAFE Research Center** (07/2024–12/2031) **ITRC (Korea Univ)**
- **(SW Star-Lab, PI) Quantum AI Empowered Second-Life Platform Technology** (07/2024–12/2031) **Star-Lab (Korea Univ)**
- *Intelligent 6G Wireless Access System Research Center* (04/2021–12/2025) **6G AI Research Center (Korea Univ)**
- *K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks* (06/2021–05/2024) **Basic Research Lab (Ajou)**
- *Nano UAV Intelligence Systems Research Lab* (10/2020–08/2023) **ADD Military Special Research Center (Kwangwoon)**
- *5G/Unmanned Vehicle Research Center (5G/UV-RC)* (06/2020–12/2022) **ITRC (Hanyang)**
- *Human Resource Development for the Biomedical Unstructured Big Data Analysis* (08/2018–12/2021) **ITRC (SNU-Hospital)**
- *Novel Data Science Driven Framework for Efficient Network Design* (06/2017–05/2020) **Basic Research Lab (Chung-Ang)**
- *Intelligent Internet of Energy (IoE) Data Research Center* (02/2020–05/2020) **ITRC (Kookmin)**

Consortium Projects

- *6GARROW: 6G AI-Native Integrated RAN-Core Networks* (09/2024–08/2027) **IITP**
- *AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development* (04/2022–12/2026) **IITP**
- *Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment* (04/2021–12/2025) **IITP**
- *Integrated Perception Technology Developments for Public Safety Platforms* (06/2019–05/2023) **NRF**
- *Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm* (07/2019–12/2021) **MHW**
- *Virtual Presence in Moving Objects through 5G (PriMO-5G)* (06/2018–06/2021) **IITP**
- *Distributed Secure Platform for Scalable Clinical OMOP CDM Models* (04/2019–12/2020) **MHW**

Individual Projects

- *Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System* (03/2025–02/2028) **NRF**
- *Research on Trainability and Learnability in Quantum Machine Learning Models* (03/2026–10/2027) **ETRI-Affiliated Institute**
- *Advancement Technology Development for Torpedo Deception Strategies in Submarines* (11/2022–11/2026) **LIG Nex1**
- *Advancement Tech Dev for Submarine Target Identification and Engagement Support Intelligence* (11/2022–11/2026) **LIG Nex1**
- *Research Planning for RL-based Scheduling in Autonomous Satellite Task Allocation* (05/2026–10/2026) **KOITA**
- *mmWave-based Liquid Classification in Next-Gen Robotic Vacuum Cleaners* (06/2026–09/2026) **LG Electronics (Robotics Lab)**
- *Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers* (09/2020–11/2025) **Samsung Electronics**
- *LEO Satellite Routing Research using Large Language Model and Reinforcement Learning* (05/2025–11/2025) **ETRI**
- *Discovering Quantum-Advantage Cases for Industry Adoption* (09/2025–11/2025) **KTL**
- *Research on Generative Quantum Machine Learning Models* (04/2025–10/2025) **ETRI-Affiliated Institute**
- *Quantum Hyper-Driving: Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility* (03/2022–02/2025) **NRF**
- *Research on Learning-based Swarm Mission Planning Algorithms* (03/2024–02/2025) **LIG Nex1**
- *Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks* (05/2024–11/2024) **ETRI**
- *Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies* (10/2023–09/2024) **NRF**
- *Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration* (12/2022–04/2024) **Hyundai Motors**
- *NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks* (09/2023–11/2023) **ETRI**
- *Routing Algorithms for LEO Satellite Networks* (12/2022–08/2023) **Solvit System**
- *Optimal Positioning Algorithms for Wide-Area Relaying Networks* (12/2022–08/2023) **Solvit System**
- *Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data* (11/2022–02/2023) **Cipherome**
- *Cellular/Wi-Fi Handover Technology Development* (02/2022–12/2022) **LG Electronics CTO Division (Smart Mobility Lab)**
- *Autonomous Intelligent COA Search Methods for Cyber-Attacks* (12/2021–11/2022) **ADD**
- *Development of Quantum Deep Reinforcement Learning Algorithm using QAOA* (10/2019–04/2022) **NRF**
- *Research Trends in Digital Twin Applications to Autonomous Driving* (03/2022–04/2022) **Hyundai NGV**
- *Distributed Learning System Design and Implementation for Clinical Applications* (02/2022–03/2022) **Cipherome**
- *mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving* (06/2019–02/2022) **NRF**
- *Research on Intelligent Agent-based CPS Security and Reliability* (04/2021–11/2021) **TTA**
- *Multi-GPU based Automotive HPC Platform Development* (04/2020–10/2020) **ETRI**
- *Super-Resolution Performance Optimization in Mobile Platforms* (05/2020–08/2020) **Samsung SDS**
- *Cooperative Deep Reinforcement Learning for Online Game Multi-Agents* (04/2020–08/2020) **ETRI**
- *Deep Learning Algorithms for mVOC Concentration Analysis* (03/2020–06/2020) **Samsung Electronics (C-Lab)**
- *Visual Recognition Software Implementation using Deep Learning Tools* (05/2019–11/2019) **Hyundai Motors**
- *Verification Testbed Implementation for Privacy-Preserving Trust Data Generation* (10/2019–11/2019) **ETRI**

- *Measurement and Analysis of Multi-Task GPU Scheduling Delays* (05/2019–10/2019) **ETRI**
- *mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services* (06/2016–05/2019) **NRF**
- *Probabilistic Decision Making and Econometric Methods for Micro-Grid* (05/2017–04/2019) **KEPCO Research Institute**
- *GPU Scheduling Performance Analysis under Queueing Delay Considerations* (05/2018–10/2018) **ETRI**
- *Improving Massive Deep Learning Training via Computation and Communication Acceleration* (04/2018–10/2018) **ETRI**
- *A Priori Techniques Research for Efficient Multi-Edge Computing* (06/2017–12/2017) **Samsung Electronics (Software Center)**
- *Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms* (04/2017–12/2017) **IITP**
- *Parsing Techniques for Artificial Neural Network (ANN) Data Processing* (09/2017–11/2017) **ETRI**

University of Southern California, Selected Projects

- *Video Aware Wireless Networks (VAWN) Research Program* **Intel Labs, Verizon Wireless, Cisco Systems**
- *60 GHz Real-Time Wireless Video Broadcasting* **Disney Research Zürich**
- *Annenberg Graduate Fellowship Award* **University of Southern California**

Korea University (KU), Selected Projects

- *Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control* (08/2025–07/2026) **PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking Univ)**
- *AI Teaching Assistant Research for LLM-based Large-Scale Education* (10/2024–02/2025) **KU University College**
- *Autonomous Mobility Control using Quantum Deep Learning* (03/2023–02/2024) **KU Insung Research Grant**
- *Mobile Access Algorithm Design using Economic Theory and AI* (03/2020–02/2021) **KU Future Research Grant**

Selected Publications

- **12,161+ Citations** (H-index: 54+, i10-index: 256+), obtained from Google Scholar Profile (as of July 18, 2026)
- **Journals and Magazines:** **174** publications; among them, **130** IEEE/Nature/APS publications

Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) *(Editors: X. Lin, J. Zhang, Y. Liu, J. Kim)*

Selected Papers

■ Top-Tier Conferences

[ICML'26] *Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Speed-Up in Query Complexity*
ICML 2026 (H. K. Lee, J. Kim, S. W. Yoon) (Acceptance Rate: 26.56%)

[ACL'26] *Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom*
ACL 2026 (J. Lee, G. Youn, S. Lee, C. Im, J. Kim, C. Yoo)

[CIKM'25] *Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models*
ACM CIKM 2025 (E. J. Roh, J. Kim) (Acceptance Rate: 30.63%)

[CIKM'25] *Filtered One-Shot Training for Quantum Architecture Search*
ACM CIKM 2025 (S. B. Son, S. Y.-C. Chen, J. Kim, S. Park) (Acceptance Rate: 30.63%)

[CIKM'25] *LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data*
ACM CIKM 2025 (G. Youn, J. Lee, J. Kim, C. Yoo) (Acceptance Rate: 30.63%)

[IPDPS'25] *AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming*
IEEE IPDPS 2025 (X. Piao, J. Shim, J. Kim, J. Kim) (Acceptance Rate: 24.71%)

[WiOpt'25] *Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach*
IEEE/IFIP WiOpt 2025 (G. S. Kim, J. Chung, T. Q. Duong, S. Park, J. Kim)

[NOMS'25] *Joint Multi-Agent Reinforcement Learning and Message-Passing for Distributed Multi-UAV Network Management using Conflict Graphs*
IEEE/IFIP NOMS 2025 (Y. Cho, H. Lee, S. Park, J. Kim)

[CIKM'24] *Hands-On Introduction to Quantum Machine Learning*
ACM CIKM 2024 (Tutorial) (S. Y.-C. Chen, J. Kim)

[WiOpt'24] *Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management*
IEEE/IFIP WiOpt 2024 (S. Lee, G. S. Kim, S. Park, J. Kim)

[CIKM'23] *Quantum Split Learning for Privacy-Preserving Information Management*
ACM CIKM 2023 (S. Park, H. Baek, J. Kim) (Acceptance Rate: 27.44%)

[CIKM'23] *Logarithmic Dimension Reduction for Quantum Neural Networks*
ACM CIKM 2023 (H. Baek, S. Park, J. Kim) (Acceptance Rate: 27.44%)

[AAAI'23] *Quantum Multi-Agent Meta Reinforcement Learning*
AAAI 2023 (W. Yun, J. Park, J. Kim) (Acceptance Rate: 19.61%)

[CIKM'22] *Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly*
ACM CIKM 2022 (W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, J. Kim) (Acceptance Rate: 29.64%)

[WiOpt'22] *Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution*
IEEE/IFIP WiOpt 2022 (M. Choi, W. Yun, J. Kim)

[INFOCOM'22] *Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks*
IEEE INFOCOM 2022 (H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim) (Acceptance Rate: 19.93%)

[ICDCS'20] *Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications*
IEEE ICDCS 2020 (Ü. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen) (Acceptance Rate: 17.98%)

[IJCAI'19] *Randomized Adversarial Imitation Learning for Autonomous Driving*
IJCAI 2019 (M. Shin, J. Kim) (Acceptance Rate: 17.89%)

[ICDCS'18] *ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture*
IEEE ICDCS 2018 (S. Ahn, J. Kim, E. Lim, W. Choi, A. Mohaisen, S. Kang) (Acceptance Rate: 20.63%)

[ACM-MM'17] *REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints*
ACM Multimedia 2017 (J. Koo, J. Yi, J. Kim, M. Hoque, S. Choi) (Acceptance Rate: 27.63%)

[MobiSys'10] *Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones*
ACM MobiSys 2010 (J. Paek, J. Kim, R. Govindan) (Acceptance Rate: 19.84%)

■ Magazines and Journals

◂ Minor ▸

[IOTJ.*minor*] USD2C: UAV-Assisted Secure Data Collection and Computation Framework for Internet of Multimedia Things (IoMT) Applications, *IEEE Internet of Things Journal*, v(n):ppp–ppp *(I. Ahmad, J. Kim, S. Shin)*

[TSUSC.*minor*] Dynamic Sun Tracking Control for Efficient Solar Power Charging in Sustainable Cube-Satellites, *IEEE Transactions on Sustainable Computing*, v(n):ppp–ppp *(G. S. Kim, S. Park, J. Kim)*

[AC.*minor*] Physics-Validated Action Masking with LLM-Guided Decomposition for Autonomous Excavation, *Automation in Construction*, v(n):ppp–ppp *(J. Cho, M. Shin, B. Kim, B. Cha, J. Kim, S. Jung)*

◂ Accepted ▸

[CIM.*accepted*] Rethinking Privacy-Preserving Distributed Learning in Quantum Hilbert Spaces, *IEEE Computational Intelligence Magazine*, v(n):ppp–ppp *(S. B. Son, J. Kim, S. Park)*

[CM.*accepted*] Sustainable 6G Non-Terrestrial Networks: Enabling Net-Zero Operation via Renewable Energy Integration and Intelligent Control, *IEEE Communications Magazine*, v(n):ppp–ppp *(H. Lee, S. Park, S. Jung, J. Kim)*

[CM.*accepted*] Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective, *IEEE Communications Magazine*, v(n):ppp–ppp *(G. S. Kim, J. Chung, S. Park, S. Jung, T. Q. Duong, J. Kim)*

[NM.*accepted*] Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks, *IEEE Network*, v(n):ppp–ppp *(H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, J. Kim)*

[INM.*accepted*] Toward Quantum Combinatorial Optimization: From QAOA Fundamentals to Industrial Deployment, *IEEE Nanotechnology Magazine*, v(n):ppp–ppp *(Y. P. Hong, J. Gil, D. Kim, S. Oh, B. S. Kim, J. Kim)*

[TMC.*accepted*] Scale-Reconfigurable Multi-Agent Reinforcement Learning for Aerial Networks in Dynamic Environments, *IEEE Transactions on Mobile Computing*, v(n):ppp–ppp *(G. S. Kim, E. J. Roh, S. Jung, S. Park, J. Kim)*

[TMC.*accepted*] Efficient Quantum Neural Architecture Search with Filtered Supernet for Mobile Learning Systems, *IEEE Transactions on Mobile Computing*, v(n):ppp–ppp *(S. B. Son, H. Ahn, S. Y.-C. Chen, S. Park, J. Kim)*

[TVT.*accepted*] Time-to-Collision Aware Autonomous Driving Motion Control under Risky Scenario Considerations, *IEEE Transactions on Vehicular Technology*, v(n):ppp–ppp *(S. Kim, J. Kim, J. Kim, S. Jung)*

◂ 2026 ▸

[FGCS'26.09] GraMA: A Gradient Matrix-Guided Assignment Method for Solving Qubit Mapping Problems, *Future Generation Computer Systems*, 182:108485 *(X. Piao, J. Kim, J.-K. Kim)*

[IOTJ'26.08] Learning-based Resilient Dynamic Routing for Fault-Tolerant Robust LEO Satellite Networks, *IEEE Internet of Things Journal*, 13(16):ppp–ppp *(G. S. Kim, C. Im, Y. G. Kim, J. Ha, J. Kim)*

[EAAI'26.08] A Comprehensive Review on Quantum Deep Neural Networks for Prognostics and Health Management: Fundamentals, Challenges and Opportunities, *Engineering Applications of Artificial Intelligence*, 177:114991 *(M. Jha, S. Chen, C. Kulkarni, J. Kim)*

[TMC'26.07] Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient Cube-Satellite Networks, *IEEE Transactions on Mobile Computing*, 25(7):10385–10401 *(S. Park, G. S. Kim, Y. Cho, S. Jung, Z. Han, J. Kim)*

[Access'26.07] Two-Stage MPC-Guided Imitation and Reinforcement Learning for Autonomous Underwater Vehicle Control in Dynamic Obstacle Environments, *IEEE Access*, 14:99636–99648 *(S. Kim, E. J. Roh, I. Song, J. Kwon, S. Kwon, Y. Kim, J. Kim)*

[IOTJ'26.06] Quantum Federated Gradient Aggregation Using the Parameter-Shift Rule, *IEEE Internet of Things Journal*, 13(11):25367–25379 *(J. Chung, C. Im, S. Park, J. Kim, W. Lee)*

[Nature'26.05] Fourier Analysis Perspective on Quantum Neural Networks, *Communications Physics (Nature)*, 9:176 *(S. Oh, E. J. Roh, A. Vasilakos, S. Park, J. Kim)*

[TVT'26.04] Large-Scale Battery-Conscious Collision-Free Path-Finding for Sustainable and Autonomous Multi-AGV Mobility Control, *IEEE Transactions on Vehicular Technology*, 75(4):5364–5375 *(Y. Cho, S. Ahn, S. Park, J. Kim)*

[TAES'26.03] Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket, *IEEE Transactions on Aerospace and Electronic Systems*, 62:3838–3863 *(G. S. Kim, J. Chung, S. Jung, S. Park, J. Kim)*

[IOTJ'26.03] Gateway-Assisted Neural Angular Routing for Hierarchical Satellite Networks, *IEEE Internet of Things Journal*, 13(6):12322–12339 *(J. Jang, I.-S. Cho, M. Shin, J. Kim, S. Jung)*

[IOTJ'26.02] Hybrid Large Language Model and Reinforcement Learning for Energy-Efficient Multisatellite Scheduling: Boosting the Performance from Scratch, *IEEE Internet of Things Journal*, 13(3):5379–5392 *(H. Ahn, G. S. Kim, I.-S. Cho, S. Jung, J. Kim)*

[PRA'26.02] Understanding Qubit Trajectories in Parametrized Quantum Circuits, *Physical Review A*, 113:022609 *(S. Oh, C. Im, S. Park, J. Kim)*

[AC'26.02] Collaborative Learning Architecture for Autonomous Excavator Planning and Execution, *Automation in Construction*, 182:106742 *(J. Cho, M. Shin, J. Kim, S. Jung)*

[TAES'26.01] Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles, *IEEE*

Transactions on Aerospace and Electronic Systems, 62:682–703 (G. S. Kim, S. Park, S. Jung, D. Mohaisen, J. Kim)

[TMC'26.01] Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks, *IEEE Transactions on Mobile Computing*, 25(1):1200–1218 (G. S. Kim, Y. Cho, J. Chung, S. Park, S. Jung, Z. Han, J. Kim)

◀ 2025 ▶

[MM'25.10-12] Quantum Jump to Virtual Worlds: High-Quality Multiple Virtual Meta-Space Realization in Metaverse, *IEEE MultiMedia*, 32(4):37–44 (S. Park, J. Kim)

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Academic Activities and Research Supervision

Research Supervision

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- **Hankyul Baek** (03/2021–02/2024 (PhD), 03/2024–03/2025 (Postdoc)), ETRI AI Safety Institute (Researcher)
- **Hyunsoo Lee** (03/2021–02/2026 (PhD)), LG Electronics (Researcher)
- **Gyu Seon Kim** (03/2023–02/2026 (PhD), 03/2026–04/2026 (Postdoc)), RealtimeVisual (Researcher, *Military Service Exception*)
- **Seok Bin Son** (03/2022–02/2027 (Expected) (PhD))
- **Emily Jimin Roh** (03/2024–02/2027 (Expected) (PhD))

■ M.S. Alumni

- **Kyeongseon Kim** (09/2017–08/2019), KAIST (Ph.D. Student in Computer Science)
- **Dohyun Kwon** (03/2018–02/2020), Hyundai Marine & Fire Insurance (Researcher)

- **Dohyun Kim** (03/2018–02/2020), Naver Webtoon (Researcher)
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■ Postdoctoral Scholars

- **Minseok Choi** (09/2018–02/2019, jointly with Prof. Andreas F. Molisch (USC)), **Kyung Hee University (Professor)**
- **Soyi Jung** (03/2021–08/2021, jointly with Prof. Marco Levorato (UC-Irvine)), **Ajou University (Professor)**
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IEEE Activities (Membership, Editorial Boards, and Services)

- *Senior Member (2018–)*, *Member (2006–2017)*
- *Associate Editor (2025–)*, **ACM Computing Surveys**
- *Associate Editor (2025–)*, **IEEE Communications Surveys and Tutorials**
- *Editor (2023–)*, **IEEE Internet of Things Journal**
- *Associate Editor (2020–)*, **IEEE Transactions on Vehicular Technology**
- *Guest Editor*, **Journal of Communications and Networks** (S.I. on Quantum Technologies for Communication Systems)
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